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Public Health Directorate

National Guidelines for Primary Health Care Physicians

Diabetes Mellitus and Metabolic Syndrome Management

Iraq

2026

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Health Care Physicians

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List of abbreviation

ACEIs	Angiotensin converting enzyme inhibitor
ARB	Angiotensin receptor blocker
BB	Beta blockers
BMI	Body mass index
BP	Blood pressure
CCBs	Calcium channel blockers
CNS	Central nervous system
CVD	Cardiovascular disease
DASH	Dietary approach to stop hypertension
DBP	Diastolic blood pressure
DKA	Diabetes ketoacidosis
DCCT	Diabetes Control and Complications Trial
DM	Diabetes Mellitus
ECG	Electrocardiography
FPG	Fasting plasma glucose
GFR	Glomerular filtration rate
HDL	High density lipoprotein
HF	Heart failure
IFG	Impaired Fasting Plasma Glucose
IFCC	International Federation of Clinical Chemistry
IGT	Impaired Oral Glucose Tolerance Test (2hr PP)
i.m	Intra muscular
i.v	Intra venous
IHD	Ischemic heart disease
ISH	International society for hypertension
JVD	Jugular vein distension
LADA	Latent autoimmune diabetes in adults
LDL	Low density lipoprotein
LVH	Left ventricular hypertrophy
MODY	Maturity –onset diabetes of the young
NGSP	National Glycohemoglobin Standardization Program
OGTT	Oral glucose tolerance test
2hr PP	Two hours post prandial
s.c	Subcutaneous
SBP	Systolic blood pressure
SMBG	Self - monitoring blood glucose
TSH	Thyroid stimulating hormone
WHO	World health organization

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Introduction

Diabetes Mellitus with its devastating complications is considered one of the leading causes of morbidity and mortality globally. It is estimated by the latest International Federation Reports (2025) that over 4 in 10 adult people with diabetes are unaware that they have the condition. Diabetes is a chronic disease that requires continuous medical care and education to prevent acute complications and to reduce the risk of long term complication. People with diabetes should receive treatment and care from a physician-coordinated team of health professionals including nurses, nutritionists, and social workers. As other countries in the region Iraq is undergoing an epidemiological transition with increasing burden of non-communicable diseases with their contributory risk factors. According to the national survey conducted in 2015, it is estimated that 13.9% of the adult Iraqi population (18+years of age), have hyperglycemia, 35.6% have high blood pressure, 33.5% are obese and 37.8% have hypercholesterolemia.

Definition

Diabetes mellitus is a metabolic disorder of multiple etiology characterized by chronic hyperglycemia with disturbances of carbohydrates, fat and protein metabolism resulting from a defect in insulin secretion, insulin resistance, or both. (WHO, 1999). It is almost always associated with multiple other metabolic abnormalities, such as hypertension, dyslipidaemia, and a pro-thrombotic state, which in combination form the Metabolic Syndrome, and greatly increase the morbidity and mortality of the disease.

Classification of Diabetes Mellitus:

Diabetes can be classified into four clinical categories, which include:

1. **Type 1 Diabetes Mellitus** – is the result of an absolute deficiency of insulin caused by an autoimmune process destroying the β islet cells of the pancreas. This is found most commonly in children and young people. Finding low insulin is nonspecific, because the level may be functionally depressed by marked hyperglycemia, dehydration or lipo toxicity. Approximately 10% of all diabetics have Type I diabetes.
2. **Type 2 Diabetes Mellitus** – this is characterized by relative resistance of body cells to insulin with progressive loss of β islet cells Insulin secretion, and is strongly associated with obesity, especially abdominal obesity. It is found primarily in older adults, but increasingly found even in obese young adults, youth and even children . Approximately 80% of all diabetics have type II diabetes, and most of these will also develop the metabolic syndrome as they age.
3. **Gestational Diabetes** – This is characterized by a relative glucose intolerance with hyperglycemia during pregnancy (during second or third trimester), and can lead to significant complications of the pregnancy, such as macrosomia and increased incidence of certain congenital defects. Although it may improve following delivery, it is also associated with an increased risk of diabetes later in life.
4. **Secondery diabetes–** these are relatively uncommon, but include various disorders of the pancreas (cystic fibrosis, chronic pancreatitis), genetic defects in pancreatic β-cell function or of insulin action, or diabetes induced by various medications (corticosteroids, anti-neoplastic drugs, anti-retroviral drugs) .

Other special forms

- Maturity-Onset Diabetes of the Young (MODY)- is the most common form of monogenic diabetes resulting from a single gene mutation. It is characterized by mild hyperglycemia, autosomal dominant inheritance, early onset of diabetes (<25 years), β- cell dysfunction with preserved insulin sensitivity and preservation of endogenous insulin secretion .

- Latent autoimmune diabetes in adults (LADA)- a disease of adults. The Immunology for Diabetes Society has specified 3 criteria for the diagnosis of LADA: Age greater than 30 years. Positive autoantibodies to islet β cells, Insulin independence for at least the initial 6 months after initial diagnosis.

Clinical Presentation

Diabetes should be suspected with any of the following symptoms:

- Polyuria (frequent urination)
- Polydypsia (thirst and frequent drinking of water or other fluids)
- Polyphagia
- Weight loss
- Intermittent blurring of the vision
- Recurrent skin infection

Diagnosis of Diabetes Mellitus

Diagnosis is established with one of four possible tests, as follows:

- Fasting plasma glucose ≥ 126 mg/dl (7.0 mmol/L) which is confirmed with a second elevated reading on a separate day. Fasting should be for minimum of 8 hours over a night.
- Oral Glucose Tolerance test (OGTT) : (ideally 75 gm anhydrous glucose dissolved in water) plasma glucose at 2 hours ≥ 200 mg/dl (11.1 mmol/L)
- Casual plasma glucose(without regard to the time since the last meal) plasma glucose ≥ 200 mg/dL (11.1 mmol/L), together with any of the above classic symptoms of hyperglycemia
- HbA1c measurement ≥ 6.5% using a standardized method of testing that is certified by NGSP (National Glycohemoglobin Standardization Program) and standardized to the Diabetes Control and Complications Trial (DCCT) assay calibrated according to International Federation of Clinical Chemistry (IFCC) standardization.

Table 1: Diagnosis of Diabetes Mellitus by Type of Test.

Test	Diabetes Values
Fasting Plasma Glucose (IFG)	≥ 126 mg/dL (≥ 7 mmol/L)[+]
Oral Glucose Tolerance Test (2hr PP) (IGT)	≥ 200 mg/dL (≥ 11.1 mmol/L)[+]
Casual Plasma Glucose	≥ 200 mg/dL (≥ 11.1 mmol/L)[++]
HbA1c	≥ 6.5%[+]

[+] In the absence of equivocal hyperglycemia, diagnosis requires two abnormal test results from the same sample or two different samples.
[++] Only diagnostic in a patient with classic symptoms of hyperglycemia or hyperglycemic crisis.

In conditions associated with an altered relationship between A1C and plasma glucose such as hemoglobinopathies, pregnancy (second and third trimesters and the postpartum period), glucose-6-phosphate dehydrogenase deficiency, HIV, hemodialysis, recent blood loss or transfusion, or erythropoietin therapy, only plasma glucose criteria should be used to diagnose diabetes

Pre-diabetes (Impaired fasting glucose and impaired glucose tolerance):

Hyperglycemia that is not sufficient to meet the diagnostic criteria for diabetes is categorized as either impaired fasting glucose or impaired glucose tolerance. (Table 1).

Table 2: Impaired Glucose Tolerance by Type of Test.

Test	Pre-Diabetes Values (Impaired glucose tolerance)
Impaired Fasting Plasma Glucose (IFG)	100- less than126 mg/dL (5.6-less than 7 mmol/L)
Impaired Oral Glucose Tolerance Test (2hr PP) (IGT)	140 –less than200 mg/dL (7.8- less than 11.1 mmol/L)
HbA1c	5.7% – 6.4%

Complications of Diabetes Mellitus

The potential long-term complications of diabetes can be roughly divided into two categories, the microvascular and the macrovascular.

- **Microvascular complications**: are primarily caused by decreased circulation through the smallest arterioles of the body's organs, and are directly associated with prolonged hyperglycemia resulting in retinopathy (leading to blindness), nephropathy (leading to renal failure), and neuropathy (leading to Charcot foot and peripheral infections). Various studies have shown that the development of these microvascular complications can be significantly delayed by strict and consistent control of the blood glucose, most commonly by maintaining glucose at less than 126 mg/dlHowever, the target should be individualized according to age and presence of co-morbidities in order to avoid the risk of hypoglycemia.

- **Macrovascular complications:** are caused by the progressive development of atherosclerosis in the major arteries of the body (aorta, carotid, coronaries and illiac vessels) leading to myocardial infarction, heart failure, stroke, and peripheral vascular disease – These complications appear to be more related to the other metabolic abnormalities that most commonly accompany diabetes, primarily hypertension, dyslipidemia, and increased thrombogenesis rather than the actual hyperglycemia alone.
  This combination of metabolic abnormalities has been termed as Metabolic Syndrome. Most studies have found that it is one or more of these macrovascular complications which is the primary cause of death in 75-80% of all diabetics.

Co-morbidities with Diabetes Mellitus

- Autoimmune Diseases (T1DM)
- Cancer
- Cognitive Impairment/ Dementia
- Fatty Liver Disease
- Pancreatitis
- Fractures
- Hearing Impairment
- HIV
- Erectile dysfunction
- Obstructive Sleep Apnea
- Periodontal Disease
- Psychosocial/Emotional Disorders

TYPE 1 DIABETES MELLITUS

Etiology:

It is caused by an autoimmune process destroying B-cell determining level of hyperglycemia associated with demonstration of anti-islet antibodies. The patient may be susceptible to other autoimmune disorders, most commonly Celiac disease and other endocrine disorders like Hashimotos thyroiditis that may occur before or after the onset of type 1 DM.

It may result from viral destruction of B cells as in congenital rubella, cytomegalovirus and Coxsackie virus infections. Although the etiological roles of viral infections are controversial.

The pubertal peak in onset of type 1 DM occurs earlier in girls than boys. This sex differences might be mediated, in part, by estrogen or by gene regulated. The pubertal changes may contribute to accelerated onset of type 1 DM in genetically susceptible females.

Diagnosis

- In the majority of young people, the presenting symptoms are usually the following:
  - Polydypsia
  - Polyphagia
  - Excessive drinking of water and fluids
  - Polyuria
  - Weight loss
  - Blurring of the vision
- Occasionally, the initial presentation may be a diabetic ketoacidosis (DKA)
- The above symptoms should be confirmed by tests for:
  - Random plasma glucose ≥ 200 mg/dl (11.1 mmol/)
  - Heavy glycosuria
  - Possible ketonuria
- Diagnostic criteria are the same for children as for adults
- Plasma glucose rather than A1C should be used to diagnose the acute onset of type I diabetes in individuals with symptoms of hyperglycemia.
- Further investigations may be required at specialized level of care to know the etiology and associated diseases as mentioned above.
- Care should be taken to avoid delay in the diagnosis and initiation of the treatment because of the risk of rapid metabolic deterioration with insulin deficiency.

Diagnostic difficulties at onset

- Infants with unexplained weight loss and hidden symptoms
- Hyperventilation of ketoacidosis misdiagnosed as pneumonia and asthma
- Abdominal pain or vomiting misdiagnosed as abdominal 'migraine' or appendicitis
- Enuresis or polyuria misdiagnosed as urinary infection
- Polydipsia misdiagnosed as habit or psychogenic drinking

The possibility of DM type 2 is to be considered among children and young people with suspected diabetes with:

- Strong family history of DM type 2
- Obese at presentation
- black or Asian family origin
- no or low insulin requirement of less than 0.5 units/kg body weight/day after the partial remission phase
- Show evidence of insulin resistance (for example, acanthosis nigricans)

Staging of Type 1 Diabetes:

Patients with criteria of stage 2 have to be referred to pediatric endocrinologist.

	Stage 1	Stage 2	Stage 3
Characteristics	• Autoimmunity • Normoglycemia • Presymptomatic	• Autoimmunity • Dysglycemia • Presymptomatic	• Autoimmunity • Overt hyperglycemia • Symptomatic
Diagnostic criteria	• Multiple autoantibodies • No IGT or IFG	• Islet autoantibodies(usually multiple) • Dysglycemia: IFG and/or IGT FPG 100-125 mg/dL (5.6-6.9mmol/L) 2-hr PG 140-199 mg/dL (7.8-11.0 mmol/L) A1C 5.7-6.4 (39-47 mmol/mol) or ≥10% increase in A1C	• Autoantibodies may become absent • Diabetes by standard criteria

Source: Classification and Diagnosis of Diabetes: Standards of Medical Care in Diabetes—2022

Management of Type 1 Diabetes Mellitus

All type 1 diabetic patients should be referred to be treated at specialized care level. It is the specialized team's task to initiate treatment in newly diagnosed patients. The general aims of the diabetes care team should be to provide:

- Multi-disciplinary care
- Recognition of potential complications
- Expert practical guidance and skill training on self-monitoring, home care and prevention of complications.
- Psychosocial support for the family

The cornerstones of management of children and adolescents with Type 1 diabetes mellitus:

1. Insulin therapy (Annex 1)
2. Monitoring of plasma glucose
3. Appropriate nutrition
4. Education of patient and family (Annex 2)
5. Establishment of good physician-patient rapport and regular follow up are essential for achieving good glycemic control.

The goals of contemporary diabetes management in children are:

1. Preventing diabetic ketoacidosis (DKA) and severe hypoglycemia
2. Establishing realistic glycemic targets and regimens adapted to individual circumstances for each child and family

3. Maintain plasma glucose and HBA1c within the normal range
4. Balancing strict glycemic control ,which reduce the risk for long term squeal and avoidance of severe hypoglycemia which is more likely with strict glycemic control
5. Maintaining normal growth and development, and emotional maturation.

Once stabilized, some of the follow-up can be done at the PHC level with the following consideration:

- Education is provided for the patients and their family members or caregivers on Insulin therapy, plasma blood glucose monitoring, effects of diet, physical activity and intercurrent illness on glycemic control, importance of adherence to management instructions, prevention and detection of hypoglycaemia, hyperglycaemia and ketosis, risk of smoking, regular eye and dental examination.
- Regular visits are associated with optimal glycemic control.
- Wearing or carrying identity of having type I diabetes may be necessary in emergency condition.

Urgent and Emergency Issues

Hypoglycemia

- Hypoglycemia is the most frequent acute complication in type I diabetes
- Clinically, hypoglycemia causes signs and symptoms of:
  - **Autonomic activation** (hunger, trembling of hands or legs, palpitations, anxiety, pallor, sweating)
  - **Neuroglycopenia** (impaired thinking, change of mood, irritability, dizziness, headache, tiredness, confusion and later convulsions and coma). This can occur without the alarming symptoms of autonomic activation, which can result in hypoglycemia of which the patient is not aware (hypoglycemia unawareness).

Grading of severity of Hypoglycemia

Mild (grade 1): Child or adolescent is aware of, responds to and self-treats the hypoglycemia. Children aged below 5–6 years can rarely be classified as grade 1 hypoglycemia because they are usually unable to help themselves.

Moderate (grade 2): Child or adolescent cannot respond to hypoglycemia and requires help from someone else, but oral treatment is successful.

Severe (grade 3): Child or adolescent is semi-conscious or unconscious or in coma with or without convulsions and may require parenteral therapy (i.m or s.c glucagon and/or i.v glucose).

- Mild hypoglycemia may cause a variety of reversible signs and symptoms characteristic of neurological dysfunction.
- Severe prolonged hypoglycemia with convulsions has the potential, particularly in young children, to cause permanent CNS impairment

Treatment of Hypoglycemia

- Mild or moderate (grade 1 or 2)
  - Immediate oral rapidly absorbed simple carbohydrate e.g.
    - 5–15 g glucose or sucrose (tablets/sugar lumps)
    - 100 ml sweet drink (glucose/sucrose drinks, syrup, juice , etc)
  - Wait 10–15 min. If no response - Repeat oral intake as above
  - As symptoms improve or normoglycemia is restored, the next meal or oral complex carbohydrate should be ingested (e.g. fruit, bread, cereal, milk)
  - Blood glucose measurements are the only way to confirm hypoglycemia if the diagnosis is uncertain. Blood glucose measurements also confirm the return of the blood glucose towards normal after hypoglycemia

- Severe (grade 3) Treatment is urgent, by available means as:

- Severe hypoglycemia with loss of consciousness with or without convulsions (particularly if there is vomiting) is most safely and rapidly reversed by injection of Glucagon, which is best given IM (or deep SC)
  - 0.5 mg for age <12 years
  - 1.0 mg for age 12+ years (or 0.1–0.2 mg/10 kg body weight)

- If glucagon is unavailable or recovery is inadequate, IV glucose should be administered slowly by trained personnel over several minutes to reverse the hypoglycemia
  With iv glucose 10–30% at a dose of 200–500 mg/kg

- If hypoglycemia is not associated with vomiting nor severe enough to remove the swallowing, spitting or gag reflexes, it is usually effective to give concentrated sugar as glucose gel/syrup/honey/jam carefully by mouth

- Referral to hospital for optimal management

Recovery phase after severe hypoglycemia

Close observation and blood glucose monitoring are essential because vomiting is common and recurrent hypoglycemia may occur. The patient will then usually require, additional oral carbohydrate and/or IV infusion of glucose (glucose 10% at 1.2–3.0 ml/kg per hour)

Prevention of Hypoglycemia

The dangerous and damaging effects of hypoglycemic episodes can be prevented primarily by the education of the patients, their parents, and other care givers, with particular attention to:

- Early warning signs and symptoms of hypoglycemia, which may be unique to the individual.
- Self-monitoring of blood glucose.
- Effects of exercise on reduction of plasma glucose, increasing the risk for hypoglycemia.
- Encouraging snacks and bedtime meals with high fiber, complex carbohydrates with consideration to glycemic index to prevent hypoglycemic attacks.
- Emergency management of hypoglycemic episodes.
- Periodic review of individual insulin management.
- Taking special care when routines are altered, such as holidays, travel or changes of season
- Repeated advice that a source of glucose or sucrose must always be immediately available
- Equipment for blood glucose measurement must be available to all young people with diabetes for immediate confirmation and safe management of hypoglycemia.
- Ideally Glucagon should be readily accessible to all parents and care givers, especially when there is a high risk of severe hypoglycemia. Training on administration of glucagon is essential.
- Children and adolescents with diabetes should wear some form of identification or warning of their disease.
- Review of glycemic targets for those at high risk (e.g. young children and those with hypoglycemic unawareness).
- Assessment of hypoglycemic episodes with the goal of learning from each episode, particularly
  - Food intake (daytime and bedtime snacks; pre- and post-exercise carbohydrate intake).
  - Insulin action profiles (e.g. rapid-acting insulin analog to reduce post-meal or nocturnal hypoglycemia; splitting evening short/rapid and bedtime intermediate-acting insulin doses).
  - Nocturnal (2.00–4.00 AM) blood glucose measurements.

Diabetic Ketoacidosis (DKA):

- Diabetic ketoacidosis is the commonest cause of diabetes –related deaths in children
- Most deaths in DKA occur as a result of cerebral edema.
- Deaths and neurological morbidity should be avoidable by reducing the incidence of DKA through:
  - Earlier diagnosis at onset.
  - Appropriate management of diabetes during inter-current illness.
  - Recognition that recurrent DKA is often caused by insulin omission.
  - Achievement of good glycemic control.

Emergency assessment to confirm the diagnosis:

- Characteristic history: polydypsia, polyuria, vomiting, anorexia, high fever.
- Clinical assessment of dehydration and acidotic breathing and consciousness level
- Biochemical confirmation of plasma glucose, glycosuria, ketonuria, PH and electrolytes.
- Early presentation of ketoacidosis should be considered keeping in mind unexplained abdominal pain, vomiting and loss of appetite

Treatment of Ketoacidosis:

Ketoacidosis in children need to be managed at hospital. A senior specialist/ consultant physician with experience in managing DKA in children should be consulted for optimal management.

Key points in treatment:

- Fluid-rehydration
- Insulin- increase needs
- Monitor ketones
- Monitor plasma glucose

TYPE 2 DIABETES MELLITUS

Type 2 diabetes mellitus frequently goes undiagnosed for years because hyperglycemia develops gradually and at earlier stages is often not severe enough for the patient to notice any of the classic symptoms of diabetes.
Patients with type 2 diabetes are generally obese, especially with central obesity which is strongly associated with a varying degree of resistance to the effects of insulin.

Metabolic Syndrome

The definition of Metabolic Syndrome is the presence of 3 or more of following risk determinants:

1- Waist circumference more than 102 cm in men, 88 cm in women.
2- Triglycerides equal or more than 150 mg/dl.
3- HDL less than 40mg/dl in men, less than50 mg/dl in women.
4- Hypertension with blood pressure equal to or more than130/85 mmHg.
5- Impaired fasting glucose equal or more than 100mg/dl.

These levels may not meet the criteria for diagnosis of diabetes or of hypertension in the normal individual, but are statistically correlated with glucose intolerance, relative insulin resistance, and a progressively higher risk of cardiovascular disease and early mortality, which has also been shown to be reduced if all of these factors are adequately controlled.

Clinical Evaluation of the Diagnosed Diabetes and/or Metabolic Syndrome:

Evaluation should be performed at the initial visit to:
• Confirm the diagnosis and the type
• Assess related risk factors
• Detect diabetes complications and comorbidity
• Develop a management plan with the patient and family

Targeted History

- Symptoms of diabetes
- Current medications and Past medical history
  - Family history of first degree relatives with a focus on diabetes, hypertension, heart and vascular problems, especially early death from cardiac causes, renal failure, autoimmune disease)
- Symptoms of diabetic complications and co-morbidities
  - Retinopathy (Vision loss or blurring)
  - Nephropathy (Peripheral edema)
  - Neuropathy (Paraesthesias or numbness of feet or hands) Cold feet
  - Intermittent cramping pain of calf or foot pain at rest, especially at night
  - Burning, tingling, or crawling sensation in feet
  - Pain and hypersensitivity of feet
  - Weakness (foot drop)

- Skin color changes (redness, cyanosis)
- Peripheral Vascular problems

- Life style factors
  - Level and type of daily exercise
  - Dietary history (Daily dietary habits, relative composition of protein, carbohydrates and fat, Type of fat taken)
  - Tobacco consumption
  - Alcohol use
- Medication and vaccination history (influenza, hepatitis, pneomococcal)
- History of gestational diabetes or delivery of large babies ≥4kg.
- Screening for psychosocial state
- Assessment of cognitive function
- Pregnancy planning

Targeted Physical Examination

- Pulse rate
- Respiratory rate
- Blood pressure (with orthostatic blood pressure measurement)
- Height and weight
  - Calculate the Body Mass Index (BMI) as below:
    BMI = Weight (kg) /Height (m^2)
    Normal BMI Value (18.5 –less than25)
- Waist circumference
  - Normal for men < 102 cm
  - Normal for women < 88cm
- Ophthalmological exam
  - Visual acuity
  - Fundoscopic exam – refer to ophthalmologist for complete examination
- Mouth and dental condition
  - Dental infections
  - Moniliasis or other mucosal abnormalities
- Thyroid abnormalities
  - Enlarged or nodular thyroid
- Cardiovascular exam
  - Apical heave, (LVH)
  - Loud S2 or S4, JVD (evidence of heart failure)
  - Arterial disease (carotid, peripheral, renal)
  - Edema

  Peripheral pulses and distal microvascular circulation
  - Dorsalis pedis and posterior tibial pulses
  - Capillary circulation in fingers and toes
- Pulmonary Exam
  - Basal crackles
  - Evidence of fluid or effusion

- Abdominal exam
  - Palpable kidney (polycystic kidney)
  - Other masses (abdominal aortic aneurysm)
  - Bruit (renal artery stenosis)
- Skin condition
  - Ischemic skin changes
  - Site of insulin injection, lipodystrophy
  - Signs of other comorbidities
- Neurological exam with a focus on:
  - Extremity deep tendon reflexes
- Foot examination:
  - Inspection of the skin, presence of foot cracks, ulcers, callous or deformities and toenails
  - Warmth and vascular assessment
  - Neurological assessment: (10-g) monofilament testing for light touch with at least one other assessment: pinprick, temperature, vibration

Standard Laboratory Evaluation

- Fasting plasma glucose
- Hemoglobin A1C
- Lipid profile (Fasting cholesterol, LDL, HDL, Triglycerides)
- Blood urea
- Serum Creatinine
- Estimated glomerular filtration rate
- Urine albumin, ketones, and glucose (dipstick)
- Urine specimen for quantitative albumin – may calculate Albumin/Creatinine ratio
- Liver function tests

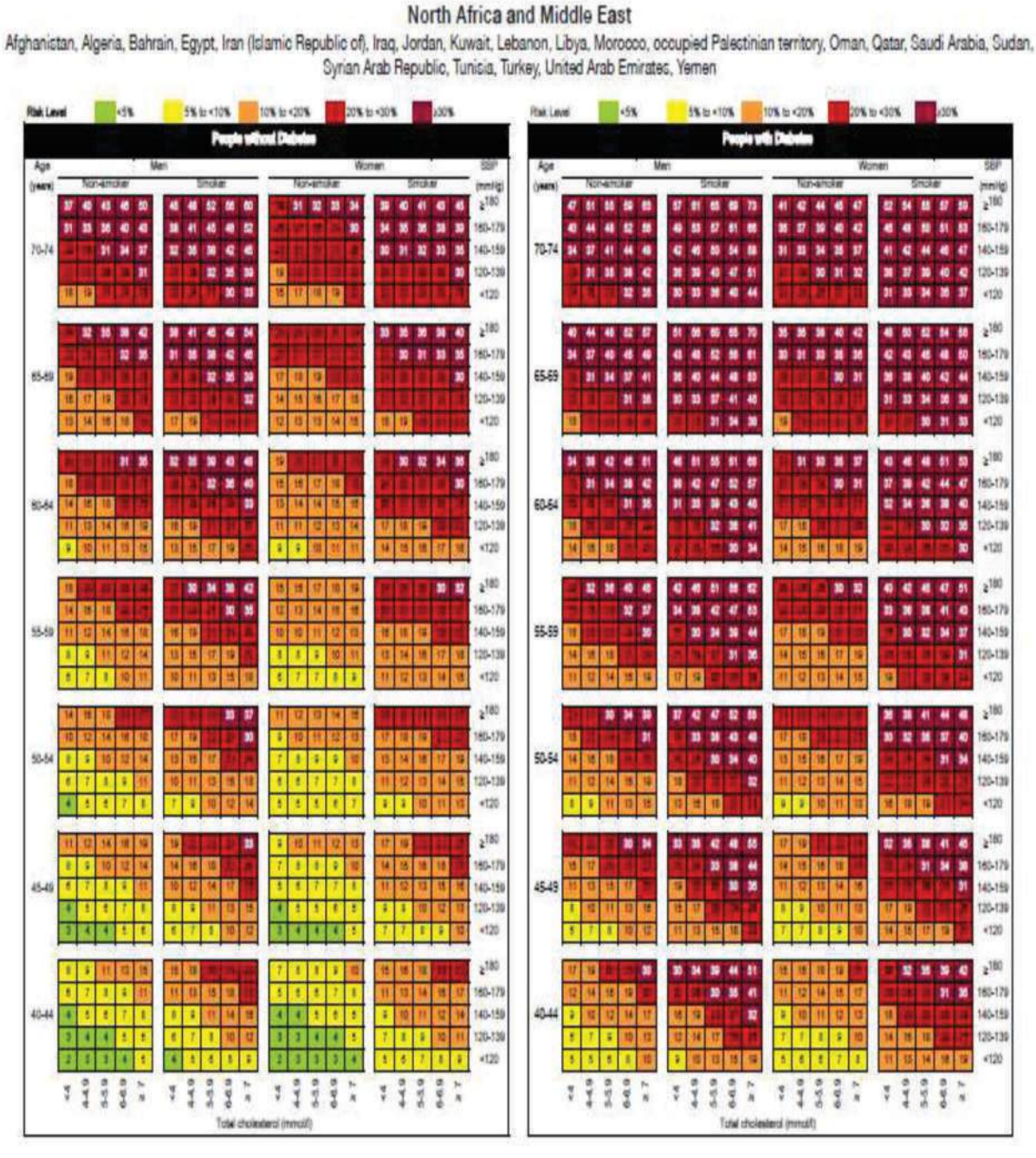
Other condition related investigations:

- TSH in type 1 diabetes
- Vitamin B12 in patients on metformin
- Serum K in patients on ACE inhibitors or ARBs or diuretics.
- ECG in adults

Cardiovascular Risk Classification:

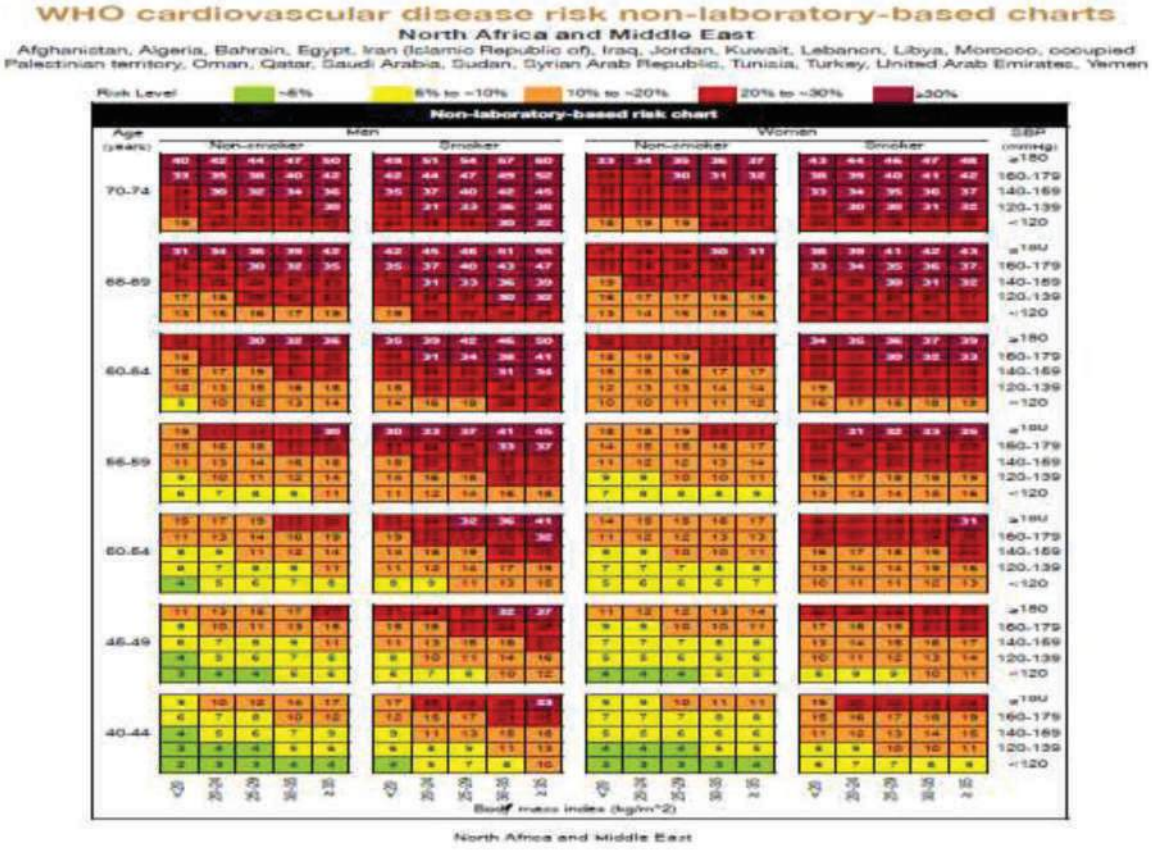
Cardiovascular risk can be expressed as the percentage chance of an individual experiencing a cardiovascular event over a pre-defined period of time, usually the next 10 years. It mainly depends on the presence of CVD risk factors commonly seen with the Metabolic Syndrome such as smoking, average blood pressure, cholesterol levels, age, and presence or absence of diabetes. The WHO-ISH chart can be used to estimate this cardiovascular disease risk over the next 10 years. There are 2 types of charts (lab.chart and non-lab. chart):

WHO- CVD Risk Chart with (LAB based chart)



Source: World Health Organization, HEARTS technical package for cardiovascular disease management in primary health care, Risk – based CVD Management; 2020.

WHO- CVD Risk Chart with (NON- LAB based chart)



Source: World Health Organization, HEARTS technical package for cardiovascular disease management in primary health care, Risk – based CVD Management; 2020.

Management of Type 2 Diabetes and the Metabolic Syndrome

It is important to recognize that each of the problems associated with diabetes, including hypertension, dyslipidemias, a pro-thrombotic tendency, and a pro-inflammatory tendency, must be aggressively and effectively treated simultaneously to reduce the morbidity and mortality of these combined metabolic abnormalities. It should be noted that the metabolic syndrome can be present even in non-obese individuals (with a BMI of < 25), and must be treated in a similar fashion.

There is a wide spectrum in the degree of insulin resistance in patients with glucose intolerance or diabetes, which reflects the severity of the other elements of the metabolic syndrome as well. These differences have resulted in a step-wise approach to the management of diabetes and the metabolic syndrome. In many cases, patients can be controlled with lifestyle changes and Step 2 management drugs. If these measures are not effective in reaching the target goals, referral to a specialist for consultation and management will be necessary.

Tight glycemic control is known to reduce microvascular complication however reduction of macrovascular complication (stroke,myocondrial infarction and peripheral vascular disease) didn't reach statistical significance but new novel drugs as GLP-I agonists and SGLT2 inhibition were found to be cardioprotective. Aggressive treatment in frail elderly and those with comorbidities were known to be harmful.

Glycemic control is not the only goal. It has also been found that the simultaneous treatment of co-existing hypertension and dyslipidemia improve the prognosis.

Goals of Diabetes and Metabolic Syndrome Management

The goals for chronic diabetes and the metabolic syndrome may be different from the goals of those who are normal or non-diabetic

Table 3 : Target Goals for Diabetes and Metabolic Syndrome according to WHO Recommendations

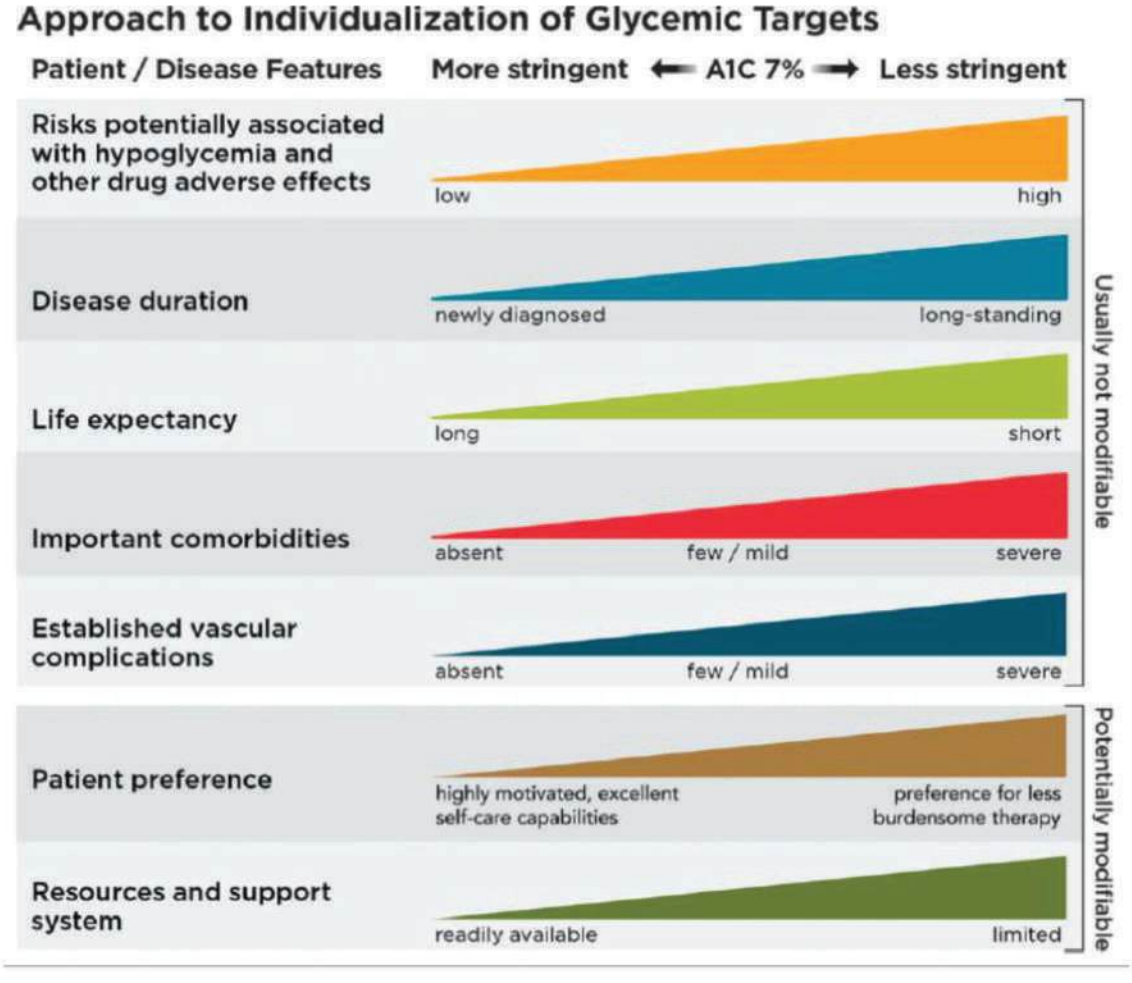
Parameter	Goal
Fasting plasma glucose	80- ≤126 mg/dl. (4.4 – ≤7.0 mmol/L)
HbA1C	(General ≤7.0%) (Elderly ≤ 8% / pregnancy ≤ 6.5%)*
Blood pressure	≤130/80 mmHg
Serum total cholesterol	<200 mg/dL. (< 5.2 mmol/L)
LDL cholesterol	<100 mg/dL. (< 2.6 mmol/L)
HDL cholesterol	Men > 40 mg/dL (>1.0 mmol/L) Women > 50 mg/dL. (>1.3 mmol/L)
Triglycerides	< 150 mg/dL. (<1.7 mmol/L)
Stop smoking	No smoking
Daily Exercise	30 min. moderate/ vigorous activity on 5-7 days .
Preprandial plasma glucose	80-130 mg/Dl
Alcohol	Absentness

NOTE – MUST WORK SIMULTANEOUSLY TOWARD ACHIEVEMENT OF ALL TARGET GOALS
***ADA 2025.**

Stepped Integrated Management of Diabetes and Metabolic Syndrome

Key points:

- Individualized approach should be adopted taking into account the potential risks of drugs adverse effects, disease duration, life expectancy, comorbidities, established vascular complications, patient preference, resources and support system.
- Diabetic with co-morbidities and/or diabetes complications should be referred for specialized care.
- In the presence of hyper catabolic state or severe osmotic symptoms (severe polyurea, polydipsia weight loss, plasma glucose≥ 300mg/dl or HbA1C ≥10%) the patient should be referred.



Source: *Standards of Medical Care in Diabetes - 2019. Diabetes Care* 2019;42(Suppl. 1):S61-S70

Glycemic control

Step1:

1.1 Life style modification (Annex 3) :

Life style intervention is essential for all diabetic and prediabetics, however pharmacotherapy should be based on HbA1c level at baseline and should be modulated according to the presence of comorbidities as chronic kidney disease , atherosclerotic cardiovascular disease and heart failure.

- **Weight loss**

Weight loss is the cornerstone of management of Type 2 diabetes and the metabolic syndrome, and is shown to result in increasing insulin sensitivity with reduction of blood pressure and dyslipidemia

- Calories restricted based on BMI and Physical activity practiced, with 45-60% in the form of complex, starchy carbohydrates, 20-35% fats, and15- 20% protein.
- Weight loss should be gradual, not more than 1 kg/week.
- Weight loss should be at least 7% of body weight.

- **Diet therapy**

The dietary therapy should include the following key modifications:

- **Weight Loss Diet**: Begin with a weight loss diet that aims to reduce LDL cholesterol, triglycerides, and blood pressure.
- **Protein Sources**:
  - Incorporate low-fat dairy products and lean meats into meals.
  - Replace most red meat with skinless chicken or fish, which are rich in omega-3 fatty acids.
- **Healthy Fats**: Use vegetable oils as the primary source of fat in cooking.
- **Sodium Intake**: the world health organization advices to limit sodium intake to less than 2.0 grams per day. This can be achieved by avoiding added salt during cooking and at the table. Salt substitutes or herbs and spices can be used in moderation for flavoring.
- **Fiber Intake**: Increase daily fiber intake to 10-15 grams by including foods such as beans, oats, other whole grains, and fiber supplements.
- **Glycemic Control:** Avoid foods with a high glycemic index, particularly those that are obviously sweet, to maintain better blood sugar levels

- **Regular exercise program**

- Regular aerobic moderate/vigorous exercise 30 minutes/day, 5 days per week is highly recommended and beneficial
- Medical examination for diabetic patients before any exercise.
- Selection of exercise according to the diabetic patients condition
- Benefits of exercise in diabetes:

- Improved insulin sensitivity.
- Improved glycemic control (in type 2 diabetes)
- Improved lipid profile (reduced triglycerides and increased HDL cholesterol)
- Lowered blood pressure
- Improved fibrinolysis.
- Potentiation of weight loss with proper diet
- Improved quality of life and self-esteem

- **Stop smoking**

  - Message to stop smoking must be clear and direct – "cutting down" is not adequate to prevent the added complications of smoking
  - Usually requires a focused behavior modification approach
  - Assist patient to specify a "stop date"
  - Offer advice on how to cope with anxiety and cravings
    - May benefit from nicotine patches or chewing gum (Nicorette)
    - Withdrawal symptoms will subside with time
    - Visit to the smoking cessation clinic

- **Sleep**

  General recommendations suggest at least 7 h of sleep per night regularly for optimal health (Annex3).

1.2 Metformin can be prescribed instantaneously with life style intervention in the first visit according to the status of the patient.

Step2:

Failure to achieve the recommended goals in 3 months after life style intervention with or without drugs require modification of antidiabetic medications. start with metformin 500 mg/day, increase as needed to 2000 mg/day in two equally divided doses

If not at target plasma glucose in response to this, Continue metformin and add new generations of sulphonylurea. If not available add glibenclamide - begin at 2.5 mg/day, increase as needed to a maximum of 15 mg/day. Glibenclamide should be avoided above the age of 60

If still not at target blood glucose in response to above: refer to specialist for consultation regarding further management.

Hypoglycemia Assessment, Prevention, and Treatment

Classification of hypoglycemia

Glycemic criteria/description	
Level 1	**Glucose <70 mg/dL (<3.9 mmol/L) and ≥54 mg/dL (≥3.0 mmol/L)**
Level 2	**Glucose <54 mg/dL (<3.0 mmol/L)**
Level 3	**A severe event characterized by altered mental and/or physical status requiring assistance for treatment of hypoglycemia, irrespective of glucose level**

Source: American Diabetes Association

Impaired hypoglycemia awareness

Impaired hypoglycemia awareness is defined as not experiencing the typical counterregulatory hormone release at low glucose levels or the associated symptoms, which often occur in individuals with long-standing diabetes or recurrent hypoglycemia. Impaired hypoglycemia awareness dramatically increases the risk for level 3 hypoglycemia.

Individuals at risk for hypoglycemia should also be screened for impaired hypoglycemia, to efficiently screen for impaired hypoglycemia awareness in clinical practice, clinicians can ask a single question based on these tools such as "Can you always feel when your blood sugar is low?" and follow up "No" responses with a more detailed evaluation.

Assessment of hypoglycemia risk among individuals treated with insulin, sulfonylureas, or meglitinides:

Clinical and biological risk factors:

Major risk factors

- Recent (within the past 3–6 months) level 2 or 3 hypoglycemia
- Intensive insulin therapy
- Impaired hypoglycemia awareness
- End-stage kidney disease

Cognitive impairment or dementia

Other risk factors

- Multiple recent episodes of level 1 hypoglycemia
- Basal insulin therapy
- Age ≥75 years
- Female sex
- High glycemic variability
- Polypharmacy
- Cardiovascular disease
- Chronic kidney disease (eGFR <60 mL/min/ 1.73 m^2 or albuminuria)
- Neuropathy
- Retinopathy
- Major depressive disorder
- Severe mental illness

Accounting for treatment plan and diabetes subtype, the oldest individuals (aged ≥75 years) have the highest risk for hypoglycemia in type 2 diabetes; younger individuals with type 1 diabetes are also at very high risk.

Intercurrent Illness

Stressful events (e.g., illness, trauma, and surgery) increase the risk for both hyperglycemia and hypoglycemia among individuals with diabetes. In severe cases, they may precipitate diabetic ketoacidosis or a nonketotic hyperglycemic hyperosmolar state, which are life-threatening.

Any individuals with diabetes experiencing illness or other stressful events should be assessed for the need for more frequent monitoring of glucose.

Clinicians should reevaluate diabetes treatment during these events and make adjustments as appropriate.

Treatment of Hypoglycemia

Health care professionals should counsel individuals with diabetes to treat hypoglycemia with fast-acting carbohydrates at the hypoglycemia alert value of 70 mg/dL (3.9 mmol/L) or less. Individuals should be counseled to recheck their glucose 15 min after ingesting carbohydrates and to repeat carbohydrate ingestion and seek care for ongoing hypoglycemia. These instructions should be reviewed at each clinical visit.

For most individuals, 15 g carbohydrates should be ingested. Individuals using automated insulin delivery systems should ingest 5–10 g carbohydrates unless there is hypoglycemia in conjunction with exercise or there has been significant overestimation of a carbohydrate/meal bolus .The acute glycemic response to food correlates better with the glucose content than with the total carbohydrate content. Pure glucose is the preferred initial treatment, but any form of carbohydrate that contains glucose will raise blood glucose. Added fat may slow and then prolong the acute glycemic response. Dietary protein intake may increase insulin secretion and should not be used to treat hypoglycemia . Ongoing insulin activity or insulin secretagogues may lead to recurrent hypoglycemia unless more food is ingested after recovery.

Prediabetes Management

Adults with overweight or obesity at high risk of type 2 diabetes are referred to an intensive lifestyle behavior change to achieve and maintain a weight reduction of at least 7% of initial body weight through healthy reduced –calorie diet and ≥ 150 min / week of moderate-intensity physical activity .A variety of eating patterns, such as Mediterranean style, intermittent fasting, and low carbohydrate, have shown benefit.

Because weight loss through behavior changes in nutrition and physical activity may not be sufficient on their own and can be difficult to maintain long term ,Pharmacologic Interventions (Metformin) to delay Type 2 Diabetes can be considered for the prevention of type 2 diabetes in adults at high risk of type 2 diabetes, especially those aged 25–59 years with BMI ≥35 kg/m^2, higher fasting plasma glucose (e.g., ≥110 mg/dL [≥6 mmol/L]), and higher A1C (e.g., ≥6.0% [≥42 mmol/mol]), and in individuals with prior gestational diabetes mellitus.

Hypertension Management

Refer to "MOH Guideline on hypertension" for complete description and protocol.

Step 1: Lifestyle modification (Annex3).

If blood pressure > 130 systolic or > 80 diastolic on 2 or more separate occasions in spite of life-style and diet changes:

Step 2:

- Classify the degree of hypertension (with revised norms based on the presence of diabetes)
- Exclude possible secondary causes of hypertension
- Begin management with ACE inhibitor medication. If not controlled add second antihypertensive medication. ACEI or ARB based treatments favorably affect the progression of diabetic nephropathy and reduce albuminuria, and ARBs have been shown to reduce

progression to macroalbuminuria., BBs, ACEIs, ARBs , CCBs and Thiazide diuretics have been shown to be beneficial in reducing CVD and stroke incidence in patients with diabetes.

- Blood pressure target should be individualized based on 10 yr cardiovascular risk and medication adverse effects. Individuals with low cardiovascular risk the target blood pressure is < 140/90 mmHg. Those with high risk the target is <130/80 if safely attained.
- Avoid combination of ACEI and ARB.

step 3

If not at target BP in response to this: Refer to specialist for management.

Dyslipidemia management

Step 1: Lifestyle modification trial (Annex3), and continue throughout management if no evidence of Atherosclerotic cardiovascular disease (ASCRD) (Annex3)

If lipid levels not at target levels in spite of life-style changes and appropriate diet and exercise:

Step 2: Begin daily statin medication (with referral to and consultation with specialist) as follows:
High intensity statin for all ages with high cardiovascular risk

- Moderate-intensity according to age:
- <40 years with additional atherosclerotic cardiovascular disease risk factors
- ≥40 years without atherosclerotic cardiovascular disease
- Statin therapy is contraindicated in pregnancy

Step 3: If lipid levels are still not at target, refer to a specialist for management

Pro-thrombotic state management if indicated

- Use aspirin therapy (75-162 mg/day) as:
- Secondary prevention strategy in those with diabetes and a history of atherosclerotic cardiovascular disease
- Primary prevention strategy in those with diabetes who are at increased cardiovascular risk, after a discussion with the patient on the benefits versus increased risk of bleeding.

Psychosocial assessment:

All patients with diabetes should have psychosocial assessment to optimize qualify of life:

- Assessment for symptoms of diabetes distress, depression, anxiety, eating disorder, and cognitive capacities using patient-appropriate standardized and validated tools at the initial visit, at periodic intervals, and when there is a change in disease, treatment, or life circumstance sharing the results with caregivers and family members.

- Screening older adults (aged ≥60 years) with diabetes for cognitive impairment and depression.

Management of Special Diabetic Patients:

- **Elderly (60 years and more)**
  - Diabetes is a highly prevalent disease in the aging population. Over one-quarter of people over the age of 65 years have diabetes, and one-half of older adults have prediabetes.
  - Diabetes management in older adults requires categorizing type of diabetes and regular assessment of medical, psychological, functional, and social domains.
  - Screening for diabetes complications in older adults should be individualized and periodically revisited, as the results of screening tests may impact targets and therapeutic approaches.
  - Older adults with diabetes are also at greater risk than other older adults for several common geriatric syndromes, such as, cognitive impairment, depression, urinary incontinence, injurious falls, persistent pain, and frailty or polypharmacy
  - Diabetes in older adults is associated with polypharmacy, increasing the risk of drugs adverse events, which necessitate revision.
  - Glycemic targets and pharmacologic treatments may need to be adjusted to minimize the occurrence of hypoglycemic events especially in those with hypoglycemia unawareness and CKD where goal must be less strict (A1c about 8%).
  - In older adults with type 2 diabetes at increased risk of hypoglycemia, medication classes with low risk of hypoglycemia are preferred.
  - Use of statin and aspirin in patients older than 75 years should be discussed with the patients to explain the benefits and adverse events thus the benefit should out weight the risk.
  - Optimal nutrition and protein intake is recommended for older adults; regular exercise, including aerobic activity, weight-bearing exercise, and/or resistance training, should be encouraged in all older adults who can safely engage in such activities.
  - Deintensification and simplification of complex treatment plans (especially insulin) is recommended to reduce the risk of hypoglycemia and polypharmacy and decrease the burden of the disease if it can be achieved within the individualized A1C target.

- **Renal disease**
  - Monitor serum creatinine and Albumin/Creatinine ratio regularly.
  - eGFR.
  - Consider adding an ACE inhibitor if Albumin/Creatinine ratio elevated, in consultation with nephrologist

- **Pregnancy**
  - Refer to Obstatrician/Gynecologist for management during pregnancy. May require transfer to insulin therapy during pregnancy

- **Coexisting heart or vascular disease**
  - Refer to cardiologist for co-management.

Follow-up Management of Diabetes and Metabolic Syndrome

All patients should be seen and evaluated on a regular basis, preferably monthly, and the following parameters should be evaluated:

Every visit:

- Fasting plasma glucose
- Blood pressure
- Medication dose review
- Patient education – may rotate topics discussed (better in groups)
- Foot examination

Every three months or as indicated:

- Weight
- Waist circumference
- Fasting glucose
- Foot exam
- Neurologic exam
    - Reflexes Especially lower extremities
    - Test for light touch, vibration or position sensation, pin prick sensation
    - May use a standardized monofilament for light touch sensation for reliability
- Medication dose review
- Patient education – may rotate topics discussed (better in groups)
- Refer for HbA1C (If not controlled or change medication until HbA1C is stable on unchanging therapy)

- **Every 6 months:**

Refer to hospital for specialized tests and specialty consultation:

- HbA1C once stable on unchanging therapy
- Lipid profile and management of dyslipidemia
- Ophthalmologic evaluation
- Quantitative albumin/creatinine ratio (request single urine specimen for quantitative albumin, and request serum creatinine)
    - Albumin/creatinine ratio calculated by dividing urine albumin (in mg/l by urine creatinine (in mg/100 ml.) Result is an empiric ratio that correlates with presence or absence of early nephropathy (normal <30, suspected nephropathy >30)
    - Refer to specialist if Albumin/Creatinine ratio >30 for follow-up. May consider starting daily dose of ACE inhibitors to preserve renal function, even if blood pressure normal or mildly elevated
- Testing for B12 every patient taking metformin

- **Every year (**especially after 5 years duration of diabetes or age >40)
    - Monitor serum cholesterol, HDL, LDL, Triglycerides
    - ECG
    - Screen for stress, anxiety, and depression

- **Foot monitoring (Annex 4):**
  - Patients with symptoms of claudication or decreased or absent pedal pulses should be referred for ankle-brachial index and for further vascular assessment as appropriate.
  - Patients who smoke or who have histories of prior lower-extremity complications, loss of protective sensation, structural abnormalities, or peripheral arterial disease should be referred to foot care specialists for ongoing preventive care and lifelong surveillance.
  - Provide general preventive foot self-care education to all patients with diabetes.

- **Counseling in family planning if appropriate**

If a woman is diagnosed with diabetes, she should consult a specialist to control her blood sugar before planning a pregnancy to prevent pregnancy complications for the mother and fetus. The American Diabetes Association recommends regulating blood sugar levels before pregnancy so that HbA1c is less than 6.5% to reduce pregnancy complications for both the mother and fetus.

Referral to a Diabetologist or other Specialists:

1. For persistent hyperglycemia or other parameters (blood pressure, lipids) uncontrolled with above treatment
2. Pregnancy and diabetes (either gestional or pre-existing diabetes) – refer to Diabetic specialist and Obstetrician
3. Initial presentation of Type 1 Diabetes
4. Diabetic ketoacidosis or hyperglycemic hyperosmolar coma refer to emergency room
5. Serious acute illness in addition to diabetes
6. Switching from oral hypoglycemic medication to insulin
7. Recognition and management of dyslipidemia
8. Presence of microvascular or macrovascular complications
9. patients who smoke or who have histories of prior lower-extremity complications, loss of protective sensation, structural abnormalities, or peripheral arterial disease are referred to foot care specialists for ongoing preventive care and lifelong surveillance.

Self-care and home care (Annex5):

Diabetes, with or without Metabolic Syndrome, is a chronic, life-long disease. The success of management depends on an effective partnership between the doctor and patient. The patient and his family members and care givers must learn about the disease, the medication and management plan, gaining skills in monitoring blood sugar, blood pressure and other elements and provide the doctor with the information to change the management as needed. Advantages include:

- Greater awareness of relationship between physical activity, diet, and medication
- Greater collaboration in management of disease
- Improved communication with health care providers

Prevention of Diabetes Mellitus :

• Screening for dysglycemia (prediabetes and type 2 diabetes):

- Testing type 2 diabetes in asymptomatic people should be considered in adults of any age who are overweight or obese (BMI ≥25 kg/m^2) and who have one or more additional risk factors for diabetes:
    - First degree relative with diabetes
    - History of cardiovascular disease
    - Hypertension or on therapy for hypertension
    - Women with polycystic ovary syndrome
- Individuals with pre-diabetes (impaired plasma glucose should be tested yearly)
- People with pre- diabetes whose condition is likely to progress, it is recommended to monitor life style modification with retest in 3-6 months.
- Women with gestational diabetes should have lifelong testing at least every 3 years. If high or impaired plasma glucose should be monitored accordingly.
- For all people, recent guidelines suggest screening to begin at age 35 years (Diabetes Care standards of care in diabetes 2025).
- The National screening system exists at age 40 years.
- If tests are normal, repeated testing should be carried out at a minimum of 3-year intervals.

• Screening for gestational diabetes

- Test for undiagnosed diabetes at the first prenatal visit in those with risk factors using standard diagnostic criteria.
- Test for gestational diabetes at 24-28 weeks of gestation in pregnant women not previously known to have diabetes.
- Test women with gestational diabetes for prediabetes or diabetes at 4-12 weeks postpartum, using the 75-g oral glucose tolerance test and clinically appropriate nonpregnancy diagnostic criteria.

• Promotion of healthy life style:

- **Diet** – Encourage low saturated fat, low sodium, high fiber diet with restricted calories
- **Physical activity:** Perform regular aerobic moderate physical activity at least 30 minutes/day, 5 days per week. Avoid sedentary life style
- **Smoking cessation**
- **Obesity management:** Keep BMI<25 and waist circumference<102cm M, <88cm F

- **Pharmacological intervention:**
  - Metformin therapy for prevention of type 2 diabetes should be considered in those with prediabetes, especially for those with BMI ≥35 Kg/M^2, those aged <60 years, and women with prior gestational diabetes mellitus.
  - This may lead to Vit B12 deficiency and periodic assessment is required.

REFERENCES

- European Society of Cardiology (ESC/EAS). 2019 Guidelines on Dyslipidaemias (Management of) ESC Clinical Practice Guidelines
  Available from: https://www.escardio.org/Guidelines/Clinical-Practice-Guidelines/Dyslipidaemias-Management-of
- International Diabetes Federation-IDF- Diabetes Atlas 11th edition 2025 Available from: https://idf.org/about-diabetes/diabetes-facts-figures/
- American Heart Association , what is metabolic syndrome?2023 Available from: https://www.heart.org/en/health-topics/metabolic-syndrome/about-metabolic-syndrome
- National health system, Metabolic syndrome, 2023 , Available from: https://www.nhs.uk/conditions/metabolic-syndrome/
- National guideline for primary health care physicians, Diabetes Mellitus and Metabolic Syndrome management, Iraq,2019.
- American Heart Association , what is metabolic syndrome?2023 Available from: https://www.heart.org/en/health-topics/metabolic-syndrome/about-metabolic-syndrome
- National Institute for health and Clinical Excellence guidelines Type 2 diabetes in adults: management, Last updated 2022, Available from: https://www.nice.org.uk/guidance/ng28/resources/type-2-diabetes-in-adults-management-pdf-1837338615493
- PIPPITT KARLY, LI MARLANAah, GURGLE HOLLY E.(2019). Diabetes Mellitus: Screening and Diagnosis. Am Fam Physician. 2016 Jan 15;93(2):103-109. Retrieved oct 31 2019
- World Health Organization: HEARTS technical package for cardiovascular disease management in primary health care, Risk – based CVD Management; 2020.Available from: https://iris.who.int/bitstream/handle/10665/333221/9789240001367-eng.pdf
- WHO. Package of Essential Noncommunicable (PEN) disease interventions for primary health care in .2020. World health organization. Available from : https://www.who.int/publications/i/item/who-package-of-essential-noncommunicable-(pen)-disease-interventions-for-primary-health-care
- American Diabetes Association (2019). Standards of Medical Care in Diabetes. American Diabetes Care. Vol 40 Suppl 1.

- NICE guideline [NG28]Published date: December 2015 Last updated: August 2019: Type 2 diabetes in adults: management. Available from https://www.nice.org.uk/guidance/ng28
- National Institute of Diabetes and Digestive and Kidney Diseases.Diabetes & Prediabetes Tests (NIH NIDDK) .2020. Available from: https://www.niddk.nih.gov/health-information/professionals/clinical-tools-patient-management/diabetes/diabetes-prediabetes
- World Health Organization (WHO) .Quitting smoking cuts your risk of developing type 2 diabetes by 30–40%.2025. Available from: https://www.who.int/news/item/14-11-2023-quitting-smoking-cuts-your-risk-of-developing-type-2-diabetes-by-30-40#:~:text=Quitting%20smoking%20cuts%20your%20risk,2%20diabetes%20by%2030%E2%80%9340%25
- American Diabetes Association Professional Practice Committee; Summary of Revisions: Standards of Care in Diabetes—2025. *Diabetes Care* 1 January 2025; 48 (Supplement_1): S6–S13. Available from: https://diabetesjournals.org/care/article/48/Supplement_1/S6/157564/Summary-of-Revisions-Standards-of-Care-in-Diabetes
- دليل تغذية مرضى داء السكري ، معهد بحوث التغذية ، دائرة الصحة العامة ، وزارة الصحة العراقية. 2024

ANNEXES

Annex 1
Insulin Therapy, for PHC Centers with the expertise and availability of the drugs.

Insulin therapy for children and young people with type 1 diabetes

The treatment of Type 1 DM is started at secondary level of care by specialized team. Insulin starting dose depends on the presentation, weather presents with classical signs and symptoms of Type 1 DM (polyuria, polydipsia, weight loss and others) which requires starting insulin only or presents with DKA that need especial protocols for treatment. It also depends on whether the patient is in prepubertal, pubertal or postpubertal.

Daily Insulin dosage varies greatly between individuals and changes over time. It therefore requires regular review and reassessment. Dosage depends on many factors such as:

- Age
- Weight
- Duration and phase of diabetes
- Nutritional intake and distribution
- Exercise patterns
- Results of plasma glucose monitoring (and glycated hemoglobin)
- Inter-current illness
- number of daily injections
- the need of flexibility in meal planning and the unique family schedule.

The optimal dose can only be determined empirically with frequent blood glucose monitoring and insulin adjustment by the team.

The doses are adjusted by the specialist according to time of action ,peak of action and duration of action which does differ from one type to another.

Types of insulin regimen are:

Currently available insulin types are classified according to duration of action into: Rapid, short, intermediate and long acting.

Representative Profiles of Insulin

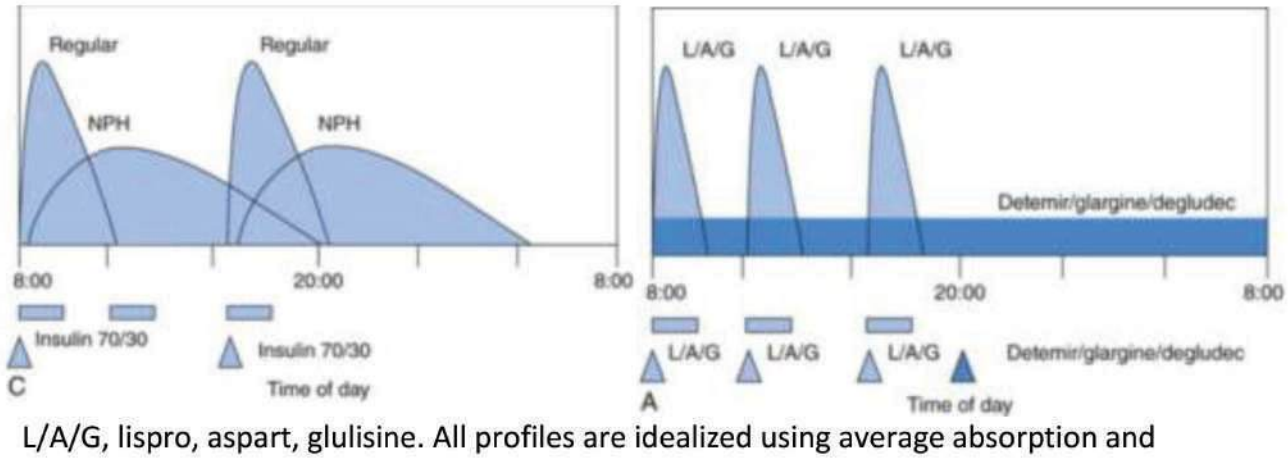
INSULIN	ONSET	PEAK ACTION	DURATION
VERY SHORT ACTING			
Lispro, aspart, glulisine	5-15 min	30-90 min	3-5 hr
SHORT ACTING			
Regular	30-60 min	2-3 hr	5-8 hr
INTERMEDIATE ACTING			
Neutral protamine Hagedorn (NPH) (isophane)	2-4 hr	4-10 hr	10-16 hr
LONG ACTING			
Glargine	2-4 hr	None	20-24 hr
Detemir	2-4 hr	6-14 hr	16-20 hr
Degludec	2-4 hr	None	24+ hr

Data from Wolfsdorf JI, ed. *Intensive Diabetes Management*. 4th ed. Alexandria, VA: American Diabetes Associations; 2009.

Type of insulin regime :

1- Conventional type of twice combination of fast acting and medium acting insulin divided into two thirds in the morning and one third in the evening.
2- Multiple injection of basal-bolus regimen comprising of long acting insulin once daily and rapid acting three times before the meals after measurement of blood glucose before the injection
3- Insulin pump therapy.

Meals are shown as rectangles below time axis. Injections are shown as labeled triangles;



L/A/G, lispro, aspart, glulisine. All profiles are idealized using average absorption and clearance rates

After the child is being discharged from hospital ,the dose of insulin should be modified keeping in mind the chance of development of honey moon period where the need of insulin drops dramatically for a variable period of time rang from few weeks up to few years thus the dose should be modified in order not to run the risk of hypoglycemia .

Key points to be considered in consultation with the specialist:

- Total daily dose of insulin need to be reduced if the child developed vomiting or diarrhea or hypoglycemia.
- The dose of insulin need to be increased during the sick days, to prevent the development of diabetic ketoacidosis and admission to the pediatric hospitals.
- Hypoglycemia need to be corrected by oral glucose syrup or subcutaneous or intramuscular Glucagon, and need to contact the specialist to correct his daily dose.
- The need for immunization of diabetic children against pneumococcal and influenza infections which are the main causes of recurrence of diabetic ketoacidosis.

Following this period the stage of complete dependence on exogenous insulin. There are two phenomenon need to think about and need specialist care:

- Somogyi phenomenon, high fasting blood glucose but blood glucose at 3-4am is low. This needs reduction of the night medium acting insulin dose.
- Dawn phenomenon, the fasting blood glucose is high and the blood glucose at 3-4am is high. This needs increase the medium acting Insulin of night dose due to increase in the antagonizing hormone to insulin like growth hormone and cortisol.

Injection sites

- The usual injection sites are
  - Front of thigh/lateral thigh (the preferred site because of ease of access, administration and for slower absorption of longer acting insulins)
  - Abdomen (the preferred site when faster absorption is required, and it may be less affected by muscle activity on exercise)
  - Buttocks (upper outer quadrant - may be useful in small children)
  - Lateral aspect of arm (in small children with little SC fat, IM injection is more likely and it may cause unsightly bruising)

Problems with injections

- Local hypersensitivity reactions to insulin injections are uncommon but when they do occur, formal identification of the insulin (or more rarely preservative) responsible may be possible with help from the manufacturer, or a trial of an alternative insulin preparation may solve the problem
- Lipohypertrophy with the accumulation of fat and fibrous tissue in lumps underneath the skin is common in children
- Lipoatrophy is now uncommon since the introduction of highly purified insulins

- Painful injections are common problems in children. Check angle and depth of injection to ensure injections are not being given IM
- Leakage of insulin is common and cannot be avoided. Encourage slower withdrawal of the needle from the skin, stretching of the skin after the needle is withdrawn, or pressure with a clean finger over the injection site
- Bruising and bleeding are more common after IM injection or tight squeezing of the skin
- Bubbles in insulin should be removed whenever possible.

FOLLOW UP OF TYPE 1 DM:-

- Frequent check of blood glucose (home readings)/ using glucometer or continuous glucose monitoring
- Check for the eye (retina)
- Check for kidney (micro/macro- albuminurea)

Annex 2
Key points in education for type 1 diabetes

Continuous education is needed with continuous review of previously learned information and addition of new concepts. It is important to take into consideration the family educational level and cultural practices. The educator must be sensitive to the age and developmental stage of the child and adolescents, and shift educational efforts from caregiver(s) to the adolescent when time is appropriate. Continued parental supervision of adolescent with diabetes is crucial to metabolic control

At initial visit or at diagnosis:

- What is type 1 diabetes cause, and how it is different from the much commoner type 2 diabetes in adult population
- Need for lifelong insulin replacement and ensuring good glycemic control
- Technique of insulin administration (should teach administration by insulin syringes and needle to all, even if pen have been prescribed), injection site rotation
- Appropriate storage of Insulin
- Targets for fasting and post prandial blood glucose
- Information on possible transient honey moon phase (Transient remission) that needs reduction of insulin to avoid hypoglycemia. The duration of this period is variable for weeks, months or rarely years.
- Brief discussion of short and long term complications
- The child can live a healthy life with normal life span if properly managed
- Self-monitoring blood glucose and maintenance of records
- Symptoms of hypoglycemia and its management
- Basic dietary advice, with a simple clear individualized diet plan

At second visit (preferably within a week after discharge or from first visit)

- Review the information given earlier, ask questions to gauge the parents understanding
- Go over the records, and reinforce action to be taken for values outside the acceptable range
- Ensure that insulin being administered is the same as has been prescribed and appropriate insulin syringe is being used (U-50syringe for 50U/ml insulin, U-100 syringe for 100U/ml insulin and see the detailed increments if it 1U to each increment or 2U to each increment)
- Teach basic modification of the dose of insulin according to plasma glucose values
- Sick day management and school-related issues, checking urinary ketones and action to be taken

At subsequent visits.

- Based on the motivation and encouragement of the patient and the family,
- Review the information given earlier
- Adherence to treatment and life style practice

Annex 3
Life Style Management

Diet Therapy for Children with Type 1 Diabetes

Importance of Nutrition: Nutrition is crucial in managing Type 1 Diabetes Mellitus (T1DM), especially during childhood and adolescence, when proper energy intake is vital for growth and development. While there are no unique nutritional needs for diabetic children apart from those required for optimal growth, individual requirements should be based on age, sex, weight, activity levels, food preferences, and cultural considerations.

Diet Composition A balanced diet for children with T1DM should consist of approximately:

- **55% Carbohydrates:** The majority should come from complex carbohydrates, such as starch, while limiting sucrose and refined sugars. Complex carbohydrates help raise blood glucose levels slowly due to their prolonged digestion and absorption.
- **30% Fats:** Total fat intake should not exceed 35%, with less than 10% coming from saturated and trans fats. It's important to replace animal fats with healthier polyunsaturated fats from vegetable sources.
- **15% Protein**

Carbohydrate Management Patients and their families should be educated about carbohydrate content in foods and how to read food labels. Carbohydrate counting and using insulin-to-carbohydrate ratios can facilitate meal planning, allowing for greater flexibility and improved quality of life. High-fiber diets are beneficial, as fiber can help manage blood glucose levels and mitigate the effects of sucrose when consumed with fiber-rich foods.

Weight Management and Meal Planning

- Regular monitoring of growth velocity, weight gain, and BMI is essential to ensure dietary plans meet the child's energy needs.
- Unexpected weight changes may indicate poor metabolic control or other health issues, necessitating further evaluation.
- Three main meals and snacks are crucial, particularly for children, pregnant women, and the elderly, to meet increased nutritional needs.

Healthy Eating Guidelines Patients should be encouraged to follow these dietary suggestions:

1. Increase intake of vegetables, whole grains, and fruits (fresh, canned, or dried).
2. Limit saturated fats and opt for healthy oils, such as olive oil.
3. Choose low-fat or fat-free dairy products and lean meats.
4. Snack on high-fiber options like unsalted nuts, fresh fruit, and raw vegetables instead of high-fat or sugary snacks.
5. Reduce salt intake by avoiding processed foods and using low-sodium alternatives.

Physical Activity Recommendations Regular physical activity is essential for diabetes management. Benefits include:

- Decreased insulin resistance.
- Improved blood sugar and A1C levels.
- Enhanced metabolic health and reduced cardiovascular risk.

The American Diabetes Association (ADA) recommends:

- **Children and Adolescents:** At least 60 minutes of moderate to vigorous aerobic activity daily, alongside muscle- and bone-strengthening activities at least three days a week.
- **Adults:** A minimum of 150 minutes of moderate to vigorous aerobic activity weekly, with resistance training sessions 2-3 times a week on non-consecutive days.

Key Considerations for Exercise

1. Individualize exercise plans based on health status and potential comorbidities.
2. Monitor blood glucose levels closely, especially if insulin is used.
3. Begin with small increments of physical activity and gradually increase frequency and intensity.

Patient Tips for Successful Management

- Track exercise and blood glucose levels.
- Stay hydrated and properly warm up and cool down during workouts.
- Dress appropriately for exercise to prevent injury and heat stress.
- Consult with a healthcare provider before starting any new exercise regimen.

By adhering to a well-structured diet and exercise plan, children with T1DM can manage their condition effectively while promoting overall health and well-being.

Lifestyle Behavior for prediabetes, type 2 diabetes

Nutrition

Nutrition counseling for weight loss in the lifestyle intervention for prediabetes and diabetes included a reduction of total fat and calories .However, evidence suggests that there is not an ideal percentage of calories from carbohydrate, protein, and fat for all people to prevent diabetes; therefore, macronutrient distribution should be based on an individualized assessment of current eating patterns, preferences, and metabolic goals. A variety of eating patterns may also be appropriate for individuals with prediabetes, including Mediterranean-style and low-carbohydrate eating plans. Observational studies have also shown that vegetarian, plant-based (may include some animal products), and Dietary Approaches to Stop Hypertension (DASH) eating patterns are associated with a lower risk of developing type 2 diabetes. Evidence suggests that the overall quality of food consumed (as measured by the Healthy Eating Index, Alternative Healthy Eating Index, and DASH score), with an emphasis on whole grains, legumes, nuts, fruits, and vegetables and minimal refined and processed foods, is also associated with a lower risk of type 2 diabetes .

Physical Activity

Moderate-intensity physical activity, such as brisk walking for 150 min/week, has shown beneficial effects in those with prediabetes. In addition to aerobic activity, a physical activity plan designed to prevent diabetes may include resistance training .Breaking up prolonged sedentary time may also be encouraged, as it lowers postprandial glucose levels .The effects of physical activity appear to extend to the prevention of gestational diabetes mellitus.

Quitting smoking

Quitting smoking is not just about healthier lungs and hearts; it's also a concrete step in reducing the risk of type 2 diabetes.

Quitting smoking not only reduces the risk of developing type 2 diabetes but also substantially improves the management and reduces the risk of diabetes complications. Evidence suggests that smoking influences the body's ability to regulate blood sugar levels, which can cause type 2 diabetes.

Smoking also increases the risk of diabetes-related complications such as cardiovascular disease, kidney failure and blindness. Smoking also delays wound healing and increases the risk of lower limb amputations.

Sleep

Sleep is highlighted as a central component in the management of prediabetes and type 2 diabetes, placing it, on the same level as other lifestyle behaviors (e.g., physical activity and nutrition). Sleep can be characterized using three key constructs: quantity, quality,

and timing (chronotype).There is now established evidence for a U-shaped association between sleep duration and type 2 diabetes incidence, with the nadir typically occurring at 7 h per day, with short (typically defined as <6 h) and long (typically defined as >9 h) sleep duration having up to a 50% increase in the risk of type 2 diabetes, including progression from prediabetes). Poor sleep quality was associated with a 40–84% increased risk of developing type 2 diabetes in a meta-analysis .Chronotype preference has been linked with many chronic diseases, including type 2 diabetes. For example, for those with a preference for evenings (i.e., going to bed late and getting up late), there was a 2.5-fold higher odds ratio for type 2 diabetes than for those with a preference for mornings (i.e., going to bed early and getting up early), independent of sleep duration and sleep sufficiency.

Annex 4
Foot Examination and Patient education

Screening symptoms of diabetic peripheral neuropathy

- Cold feet
- Intermittent cramping pain of calf or foot pain at rest, especially at night
- Burning, tingling, or crawling sensation in feet
- Pain and hypersensitivity of feet
- Weakness (foot drop)
- Skin color changes (redness, cyanosis)

Screening test for high risk of neuropathy

- Use 10 gm. Monofilament to test for light touch
- Apply Monofilament to sole of foot distally sufficient to bend monofilament – ask patient if and where pressure is felt
- Inability to feel 10 gm. pressure is high risk for future skin damage and ulceration

Patient education to prevent foot injury

- Shoes: Should be low heeled, made of soft leather or fabric, neither tight nor loose, with wide toes to avoid pressure, and the arch of shoe filling properly with the arch of the foot. Proper arch inserts should be used. Advise patient to look carefully to the inside of shoes before wearing to avoid pressure points from shoe

- New Shoes: Should be neither tight nor loose. Should be worn only for 2 hours on the first day, then increase daily use by one hour until shoes become comfortable

- Stockings and socks: Better made of cotton, thick, and warm with loose garters

- Bare foot walking: Absolutely prohibited whether indoors or outdoors

- Management of dry skin and cracks:
  - Dry skin without cracks: Soak feet daily in warm tap water for 10 – 15 minutes, dry gently, and rub with mineral oil or thick moisturizing cream to keep moisture.
  - Dry skin with cracks: Rub callus at edges of cracks with file or rough stone. Then soak feet in mild soap water for 10 – 15 minutes. Cover cracks with antiseptic such as neomycin ointment – and rub feet with mineral oil.

- Management of corns and calluses:
  - To remove excess corns or calluses ask patient to soak feet in warm tap water with mild soap for 10 minutes, then rub excess tissues by a file. If not effective consult podiatrist. sharp object use is not allowed.

- Nail care:
  - Clean around nails with a wood stick. If nails are long, file them. Filing should be straight and not shorter than the underlying soft tissues of the toes.
  - Soak brittle nails for 30 minutes each night in warm tap water

- Care of abrasions and minor trauma and infection

- Patient should consult his physician, even in case of minor injuries. If a physician is not easily accessible at all times, patients should consult with a physician in case of redness, blistering or swelling
- Cover area with sterile gauze fitted by non-adhesive plaster. Advise patient not to use limb excessively and to elevate foot while sitting.
- Avoid irritant antiseptics
- Treat infections aggressively

Home monitoring and care for feet:

It is important for each diabetic patient to develop a habit of visually inspecting both feet at least once weekly, or more often if the diabetes is advanced. The elements of this inspection should include the following:

- Daily washing and inspection
- Keeping walking areas clear of dangerous objects
- Appropriate footwear (selection, fitting and use)
- Using slippers indoors-no bare feet allowed at any time!
- Avoid extreme temperatures of either cold or heat on the feet
- Avoid soaking of the feet for long periods of time, which can lead to burns and excessive softening of the skin
- Report foot problems promptly to a physician (infections, ulcers, and cuts that do not heal)

Annex 5
Self-monitoring and Home Care of Diabetes and Metabolic Syndrome

The important elements are the following:

- Life style modification (annex 3)
- Know and be alert to danger signs of diabetes and the metabolic syndrome
  - Hypoglycemia symptoms – light-headedness, difficulty walking, confusion, disorientation
    - The treatment for hypoglycemia (15 grams of sugar), which should be carried with most diabetics
    - Actions to take to prevent relapse of hypoglycemia – regular scheduled eating, avoid fasting, balance food intake with exercise, confirm with doctor the dosages of medication
  - Possible vaginal yeast infections in women with diabetes
  - Postural dizziness with changes in position, especially for those with metabolic syndrome and several controlling medications
  - Any suggestion of a serious complication:
    - chest pain or pressure
    - weakness on one side of the body
    - vision changes
    - loss of consciousness, even if only for a few minutes
- Self/ home blood glucose monitoring
  - Patients should be encouraged to purchase a home glucose monitor, and taught how to use it effectively. Blood glucose results can then be recorded in a logbook with the date and any pertinent observations, for review by the monitoring physician
  - Home glucose monitoring is especially helpful for those patients who are unstable, whose diet is erratic, and who are on insulin.
  - Ideally monitoring should be performed 4-6times daily: fasting, Before and 2- hours after meals and during night. Also checked during and few hours after vigorous sport or exercise; to confirm presence of hypoglycemia and documentation of recovery. More frequent checking is needed on sick days.
  - Since each glucose monitor requires a unique set of test strips, it is important that the patient purchase a monitor for which the test strips are easily and economically available in his/her region.
  - The doctor or nurse should review the use of the glucose monitor with the patient to be sure that the correct technique is followed for accurate results. The frequency of recommended use of the monitor will depend on the stability of the patient's diabetes, the age, the medications being used (especially if insulin is used), and whether or not the patient is susceptible to episodes of hypoglycemia

- Home blood pressure monitoring
  - Since the majority of those with adult onset diabetes also have increased blood pressure, it is very helpful for the patient to also purchase a blood pressure apparatus to self-monitor the blood pressure in varying stages of life. This will assist the doctor in deciding on the proper medications and dosage for best control of the blood pressure

- Monitoring of blood lipids
  - Most adult patients with diabetes also have abnormal blood lipids (fats), which increase the risk of atherosclerosis and problems such as heart attacks and stroke. These should be monitored on a regular basis, at least once or twice each year, to help the doctor decide what treatments should be used to help control the lipids. This is one element of care that must be done by a physician or hospital laboratory.

- Home monitoring and care of the feet (annex 4)

Annex 6
Performance Checklist: Diabetes Mellitus type II and Metabolic Syndrome

Task	Achieved?		Comments
	Yes	No	
History: patient is asked about:			
Personal, family, and past history			
Symptoms related to diabetes			
Symptoms of coexisting illness(hypertension, liver disease, heart disease, high cholesterol)			
Frequency of acute complications (DKA – hypoglyceamia			
Full dietary history(times of meals, weight changes)			
Current medications used for coexisting diseases(steroids, thiazides)			
Methods of glucose monitoring			
Physical Examination			
Height & weight and waist circumference			
Heart rate, blood pressure			
Palpate peripheral pulses			
Examine feet for deformities, cracking, infection, ulcers			
Examine mouth, teeth, and gums			
Examine thyroid gland			
Examine skin(infection, site of insulin injection			
Chest & heart			
Abdomen (liver, spleen & loin)			
Neurological Examination (Vibration sense, peripheral numbness, ankle jerk)			
Patient Education			
Uses simple clear language			
Periodically checks to confirm if patient understands instructions			
Ask patient if he /she has any questions			
Educational messages			
Basic pathophysiology of diabetes and metabolic synd.			
Nutrition(caloric requirement, food types)			
Drugs: all drugs for diabetes/metabolic syndrome			

Task	Achieved?		Comments
	Yes	No	
Exercises: proper methods & timing			
Glucose and blood pressure monitoring			
Hypoglycemia: counsel about symptoms & prevention			
Management of other illnesses			
Long term complications and how can be prevented			
Personal hygiene			
Foot care			
Referral of patients to educational session (nutritionist, nurse specialist for diabetes)			
Monthly Diagnostic tests/procedures: Records the following tests/procedures on a monthly basis			
Fasting plasma glucose(FPG)&/or2H PPPG			
Urine glucose and ketones			
Quarterly Diagnostic tests/procedures:			
Fasting plasma glucose			
HbA1C			
Orders/records following tests every 6 months:			
HbA1C			
Lipid profile			
Quantitative Albumin/Creatinine ratio			
B12			
Annual Diagnostic tests/procedures:			
Serum Cholesterol, HDL, LDL, Triglycerides			
Retinal examination (ophthalmologist referral)			
Serum creatinine			
ECG (patients > 40 years)			
Medication			
Appropriate drug prescription according to guidelines			
Referral			
Appropriate referral for consultation according to guidelines			

Annex (7): Medication for lowering glucose in type 2 diabetes

Medication (route of administration)	Glucose-lowering efficacy	Hypoglycemia risk	Weight effects	CV effects		Kidney effects		MASH effects	Clinical considerations and adverse effects
				Effect on MACE	Effect on HF	Progression of CKD	Dosing/use considerations*		
Metformin (oral)	High	No	Neutral (potential for modest loss)	Potential benefit	Neutral	Neutral	Contraindicated with eGFR <30 mL/min/ 1.73 m²	Neutral	GI side effects: mitigate with slow dose titration, extended-release formulations, and administration with food. Potential for vitamin B12 deficiency: monitor and replete as appropriate.
SGLT2 inhibitors (oral)	Intermediate to high	No	Loss (intermediate)	Benefit: canagliflozin, empagliflozin	Benefit: canagliflozin, dapagliflozin, empagliflozin, ertugliflozin	Benefit: canagliflozin, dapagliflozin, empagliflozin	-See labels of individual agents for dosage considerations for kidney function • Glucose-lowering effect is minimal at eGFR <45 and mL/min/1.73m² lower; continue for cardiovascular and kidney benefit until dialysis or transplantation	Unknown	-DKA risk in individuals with insulin deficiency (rare in T2D): discontinue, evaluate, and treat promptly if suspected; be aware of predisposing risk factors and clinical presentations (including euglycemic DKA); mitigate risk with sick-day planning; discontinue before scheduled surgery (e.g., 3-4 days), during critical illness, or during prolonged fasting. -Genital mycotic infections: mitigate risk with genital hygiene and avoid use in high-risk individuals. - Necrotizing fasciitis of the perineum (Fournier gangrene): rare; prompt treatment if suspected. -Intravascular volume depletion: attention to volume status and blood pressure, particularly when ill or fasting; adjust other volume-contracting agents as applicable; monitor kidney function upon initiation.

Annex (7): Medication for lowering glucose in type 2 diabetes (continue)

Medication (route of administration)	Glucose-lowering efficacy	Hypoglycemia risk	Weight effects	CV effects		Kidney effects		MASH effects	Clinical considerations and adverse effects
				Effect on MACE	Effect on HF	Progression of CKD	Dosing/use considerations*		
Pioglitazone (oral)	High	No	Gain	Potential benefit	Increased risk	Neutral	No dose adjustment required Generally not recommended in kidney impairment due to potential for fluid retention	Potential benefit	• Increased risk of HF and fluid retention Do not use in the setting of HF • Risk of bone fractures • Bladder cancer do not use in individuals with active bladder cancer and use caution in those with prior history of bladder cancer
Sulfonylureas (2nd generation) (oral)	High	Yes	Gain	Neutral	Neutral	Neutral	Glyburide generally not recommended in CKD Glipizide and glimepiride : initiate conservatively to avoid hypoglycemia	Unknown	• FDA Special Warning on increased risk of CV mortality based on studies of an older sulfonylurea (tolbutamide) glimepiride shown to be CV safe (see text) • Use with caution in individuals at risk for hypoglycemia particularly if in combination with insulin

Annex (7): Medication for lowering glucose in type 2 diabetes (continue)

Insulin (human) (SQ regular insulin also available as inhaled formulation) Insulin (analogs) (SQ)	High to very high	Yes	Gain	Neutral	Neutral	Neutral	Lower insulin doses required with a decrease in eGFR titrate per clinical response	unknown	• Injection site reactions ·Higher risk of hypoglycemia with human insulin (NPH or premixed formulations) vs analogs· • Risk of hypoglycemia and duration of activity increases with the severity of impaired kidney function • Refer to device-specific instructions for insulins compatible with different delivery systems (ie pumps connected insulin pens insulin patches)

Source: American Diabetes Association

chronic kidney disease; CV, cardiovascular; DKA, diabetic ketoacidosis; DPP-4, dipeptidyl peptidase 4; eGFR, estimated glomerular filtration rate; FDA, U.S. Food and Drug Administration; GI, gastrointestinal; GIP, glucose-dependent insulinotropic polypeptide; GLP-1 RA, glucagon-like peptide 1 receptor agonist; HF, heart failure; MACE, major adverse cardiovascular events; MASH, metabolic dysfunction-associated steatohepa- titis; SGLT2, sodium-glucose cotransporter 2; SQ, subcutaneous; T2D, type 2 diabetes. *For agent-specific dosing recommendations, please re- fer to manufacturers' prescribing information. 1Tsapas et al. (106). 2Tsapas et al. (241). Adapted from Davies et al. (89).

TYPE 2 DIABETES MANAGEMENT PROTOCOL

Work up of patient with type 2 Diabetes

FPG ≥ 126 mg/dl (7 mmol/L) or RPG ≥ 200 mg/dl (11.1 mmol/L)

Life style modification:
Counsel on diet and physical activity
+/-
Metformin 500mg

Review in 3 months
If goal not achieved escalate metformin to 1000 -1500 mg in divided doses
Counsel on diet and physical activity and adherence

Review in 3 months
If goal not achieved re-escalate dose to 2000 mg in divided doses
Counsel on diet and physical activity adherence

Review in 3 months
If goal not achieved, ADD Sulphonylurea (Glimepiride OR glibenclamide OR Gliclazide) according to potency and risk of hypoglycemia
Counsel on hypoglycaemia at all subsequent visits

Review in 3 months
If goal not achieved escalate gradually to maximum dose

Review in 3 months If goal not achieved, despite adherence to medication, healthy diet and physical activity, refer to higher-level health care facility

The patient should be referred

- In the presence of hyper catabolic state or severe osmotic symptoms (severe polyurea, polydipsia, weight loss loss, plasma glucose≥ 300mg/dl or HbA1C ≥10%) the patient should be referred.
- Diabetic with co-morbidities and/or diabetes complications

Glibinclamide Starting dose 2.5 mg/day, escalate gradually to a maximum of 15 mg/day
Glimepiride Staring dose 1mg, escalate gradually to a maximum of 4 mg/day
Gliclazide starting dose 40-80 mg 1x daily increase to 80 mg x2 daily.

تصميم وطبع
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